

Prob. 1: For the questions 1,2,3,4,5 consider the circuit shown in figure 1 and answer (use nodal analysis method):

1) The value of V is: -

- a) -200mV
b) 200mV
c) 800mV
d) 400mV

2) The value of the current i is:

- a) 2.8A
b) 1.4A
c) -0.7A
d) 0.7A

3) The power absorbed by 6Ω resistor is:

- a) -10.46W
b) 10.46W
c) 5.23W
d) 6.46W

4) The power absorbed by 6V voltage source is:

- a) -22.4W
b) 44.8W
c) 11.2W
d) -44.8W

5) Write the required equations (do not simplify)

$$\frac{V_1 - 14}{4} = \frac{V_1}{3} + \frac{V_2}{6} + \frac{V_2}{2}$$

$$V_2 - V_1 = 6$$

$$i = \frac{V}{2}$$

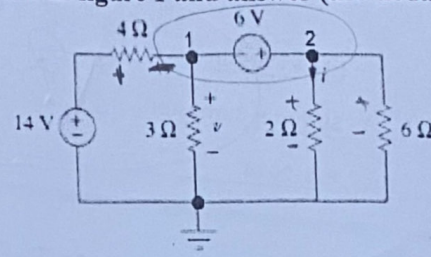


Fig.1

Prob. 2: Consider the circuit shown in figure 2. Use superposition theorem and answer:

a. The current I due to the 12V voltage source is:

$$I' = 1.277 \text{ A}$$

b. The current I due to the 2A current source is:

$$I'' = 1.333 \text{ A}$$

c. The current I due to both sources is:

$$I = I' + I'' = 2.6 \text{ A}$$

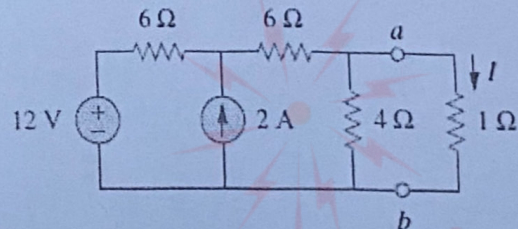


Fig. 2

Prob.3 : (a) The charge transferred through a circuit element is given by $q(t) = 4t^2 + 3t$ C. Calculate the current entering the element at instant $t = 4$ s.

$$i(4) = 35 \text{ A}$$

(b) A load draws 500 W when connected to a 230 V line. Calculate - the energy consumed in 3 hours and the conductance of the load?

$$W = 5.4 \text{ MJ}$$

(c) Refer to figure 3; write the mesh equations in matrix form:

$$\begin{bmatrix} 9 & -2 & -2 & 0 & 0 \\ -2 & 10 & -4 & -1 & -1 \\ -2 & -4 & 9 & 0 & 0 \\ 0 & -1 & 0 & 8 & -3 \\ 0 & -1 & 0 & -3 & 4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \\ V_5 \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \\ -6 \\ 0 \\ -6 \end{bmatrix}$$

Good luck

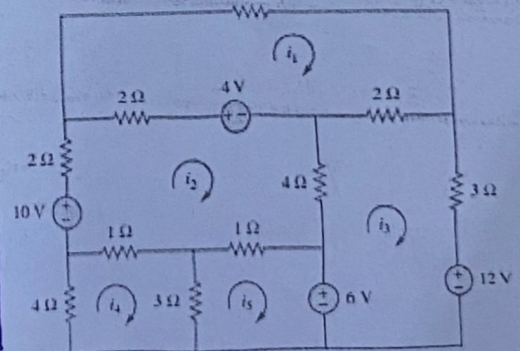
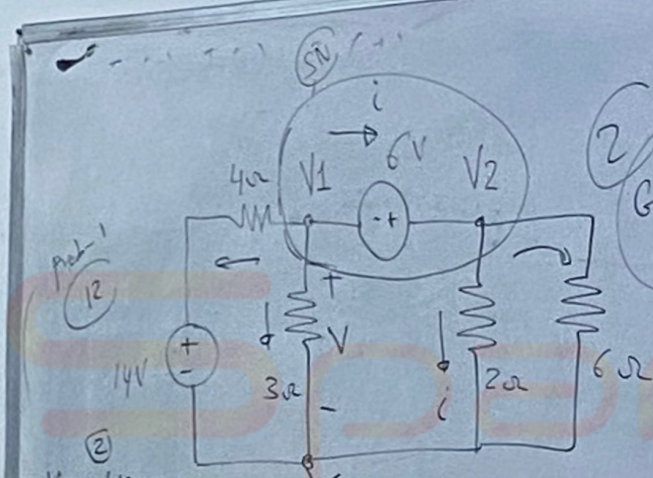


Fig 3.



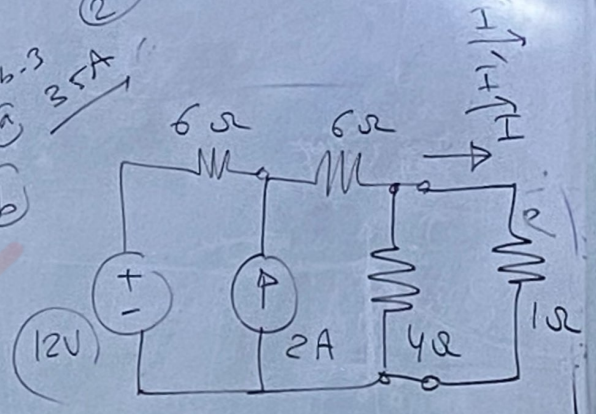
- ① $V = -400 \text{ mV}$
- ② $i = 2.8 \text{ A}$
- ③ $P = 5.23 \text{ W}$
- ④ $P = -22.4 \text{ W}$

$$\frac{V_1 - 14}{4} + \frac{V_1}{3} + \frac{V_2}{2} + \frac{V_2}{6} = 0 \quad \text{--- ①}$$

$$V_2 - V_1 = 6 \quad \text{--- ②}$$

$v = f(t)$
 $i = f(t)$
 $G = 9.452 \text{ mS}$
 5Ω
 1500 Wh
 1.516 Wh
 $5.4 \times 10^6 \text{ J}$

Prob-3
 a) 3.5 A
 b)



$$I = I' + I''$$

$$I' = 0.75 \text{ A} \quad \text{③}$$

$$I'' = 0.75 \text{ A} \quad \text{⑤}$$

$$I = 1.5 \text{ A} \quad \text{⑦}$$